

American College of Radiology
ACR Appropriateness Criteria®

Right Upper Quadrant Pain

Variant 1: Right upper quadrant pain. Unknown etiology. Initial Imaging.

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations									
							1	2	3	4	5	6	7	8	9	
US abdomen	Usually appropriate	Moderate	O 0 mSv	O 0 mSv [ped]	9	9	1	0	0	0	0	0	0	1	4	10
		References	Study Quality													
		16 (15256633)	4													
		7 (26526429)	4													
		25 (22798223)	Good													
		28 (11420481)	3													
		26 (31493199)	3													
		27 (28586152)	4													
		29 (15208132)	4													
CT abdomen with IV contrast	Usually appropriate	Limited	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	8	8	0	0	1	0	0	0	7	7	1	
		References	Study Quality													
		7 (26526429)	4													
		10 (25424561)	3													
		9 (29702020)	3													
CT abdomen without IV contrast	May be appropriate	Limited	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	5	5	1	0	0	5	6	1	2	1	0	
		References	Study Quality													
		7 (26526429)	4													

		15 (14661666)	4													
CT abdomen without and with IV contrast	Usually not appropriate	Limited	☢☢☢☢ 10-30 mSv	☢☢☢☢☢☢ 10-30 mSv [ped]	2	2	4	9	2	0	0	1	0	0	0	
		References	Study Quality													
		7 (26526429)	4													
		10 (25424561)	3													
		9 (29702020)	3													

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations								
							1	2	3	4	5	6	7	8	9
US abdomen	Usually appropriate	Strong	0 0 mSv	0 0 mSv [ped]	9	9	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		40 (7255722)		3											
		16 (15256633)		4											
		43 (12467321)		3											
		17 (19098200)		4											
		41 (7730462)		3											
		15 (14661666)		4											
		42 (7979854)		4											
		22 (10749232)		2											
		7 (26526429)		4											
		30 (24224584)		4											
		44 (26070243)		4											
		CT abdomen with IV contrast	May be appropriate	Strong	☢☢☢ 1-10 mSv										☢☢☢☢ 3-10 mSv [ped]
		References		Study Quality											

16 (15256633)	4
31 (11804880)	3
17 (19098200)	4
15 (14661666)	4
33 (19559349)	2
34 (20489092)	4
32 (16671375)	3
35 (19497464)	3
7 (26526429)	4
30 (24224584)	4
10 (25424561)	3

MRI abdomen without and with IV contrast with MRCP	May be appropriate	Limited	0 0 mSv	0 0 mSv [ped]	6	6	0	0	0	0	0	0	0	0	0
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References	Study Quality
36 (19711404)	4
37 (17581902)	3
38 (14628868)	2
7 (26526429)	4

MRI abdomen without IV contrast with MRCP	May be appropriate	Limited	0 0 mSv	0 0 mSv [ped]	5	5	0	0	0	0	0	0	0	0	0
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References	Study Quality
36 (19711404)	4
37 (17581902)	3
38 (14628868)	2
7 (26526429)	4

HIDA scan	May be appropriate	Limited	☢☢ 0.1-1mSv		5	5	0	0	0	0	0	0	0	0	0
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References	Study Quality
40 (7255722)	3
16 (15256633)	4
17 (19098200)	4

Variant 3: Right upper quadrant pain. No fever and no high white blood cell (WBC) count. Suspected biliary disease. Negative or equivocal ultrasound. Next imaging study.

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations								
							1	2	3	4	5	6	7	8	9
MRI abdomen without and with IV contrast with MRCP	Usually appropriate	Limited	O 0 mSv	O 0 mSv [ped]	8	8	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		38 (14628868)		2											
		7 (26526429)		4											
CT abdomen with IV contrast	Usually appropriate	Limited	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	7	7	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		7 (26526429)		4											
		10 (25424561)		3											
MRI abdomen without IV contrast with MRCP	Usually appropriate	Limited	O 0 mSv	O 0 mSv [ped]	7	7	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		38 (14628868)		2											
		7 (26526429)		4											
HIDA scan	May be appropriate	Limited	☼☼ 0.1-1mSv		5	5	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		45 (23731969)		4											
CT abdomen without IV contrast	May be appropriate	Limited	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	4	4	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		7 (26526429)		4											
		10 (25424561)		3											

		34 (20489092)		4													
		32 (16671375)		3													
		35 (19497464)		3													
		46 (22332066)		2													
		7 (26526429)		4													
		30 (24224584)		4													
		10 (25424561)		3													
		HIDA scan	Usually appropriate	Moderate	☼☼☼ 0.1-1mSv												
		References		Study Quality													
		42 (7979854)		4													
		25 (22798223)		Good													
MRI abdomen without IV contrast with MRCP	May be appropriate	Strong	○ ○ mSv		○ ○ mSv [ped]	6	6	0	0	0	0	0	0	0	0	0	
		References		Study Quality													
		49 (26526902)		3													
		36 (19711404)		4													
		37 (17581902)		3													
		47 (10937751)		2													
		38 (14628868)		2													
		48 (9676460)		2													
		7 (26526429)		4													
		25 (22798223)		Good													
CT abdomen without IV contrast	May be appropriate	Strong	☼☼☼ 1-10 mSv		☼☼☼☼ 3-10 mSv [ped]	4	4	0	0	0	0	0	0	0	0	0	
		References		Study Quality													
		16 (15256633)		4													
		31 (11804880)		3													
		17 (19098200)		4													
		15 (14661666)		4													
		33 (19559349)		2													

		38 (14628868)			2										
		7 (26526429)			4										
CT abdomen without IV contrast	May be appropriate	Limited	☢☢☢ 1-10 mSv	☢☢☢☢ 3-10 mSv [ped]	4	4	0	0	0	0	0	0	0	0	0
		References			Study Quality										
		31 (11804880)			3										
		50 (20489669)			3										
		7 (26526429)			4										
		10 (25424561)			3										
CT abdomen without and with IV contrast	Usually not appropriate	Limited	☢☢☢☢ 10-30 mSv	☢☢☢☢☢ 10-30 mSv [ped]	3	3	0	0	0	0	0	0	0	0	0
		References			Study Quality										
		31 (11804880)			3										
		50 (20489669)			3										
		7 (26526429)			4										
		10 (25424561)			3										

Appendix Key

A more complete discussion of the items presented below can be found by accessing the supporting documents at the designated hyperlinks.

Appropriateness Category: The panel's recommendation for a procedure based on the assessment of the risks and benefits of performing the procedure for the specified clinical scenario.

SOE: Strength of Evidence. The assessment of the amount and quality of evidence found in the peer reviewed medical literature for an appropriateness recommendation.

- **References:** The citation number and PMID for the reference(s) associated with the recommendation.
- **Study Quality:** The assessment of the quality of an individual reference based on the number of study quality elements described in the reference.

RRL: Relative Radiation Level. A population based assessment of the amount of radiation a typical patient may be exposed to during the specified procedure.

Rating: The final rating (1-9 scale) for the procedure as determined by the panel during rating rounds.

Median: The median rating (1-9 scale) for the procedure as determined by the panel during rating rounds.

Final tabulations: A histogram showing the number of panel members who rated the procedure as noted in the column heading (ie, 1, 2, 3, etc.).

Additional supporting documents about the AC methodology and processes can be found at www.acr.org/ac.